Autonomous Multi-Robot System for Dynamic Warehouse Operations

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Abstract ($\star\star\star$)

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This project investigates the design, rigorous mathematical formulation, and extensive simulation of a centralized control framework for a Multi-Robot System (MRS) operating within a stochastic, dynamic warehouse environment. The primary engineering challenge addressed is the heterogeneous task allocation problem, where a continuous, stochastic stream of tasks necessitates either single-agent execution or cooperative multi-agent transport. Unlike static allocation problems often explored in the literature, this system must handle an infinite horizon of tasks, requiring a robust and continuous re-planning architecture.

We utilize a hybrid modeling approach that combines discrete event systems for task allocation with continuous dynamical systems for motion control. The specific properties analyzed include the asymptotic stability of the cooperative formation controller and the safety guarantees provided by the collision avoidance logic. The analysis techniques applied include Lyapunov's Direct Method for stability proof and Artificial Potential Field (APF) theory for safety verification.

To validate our theoretical findings, we developed a comprehensive simulation environment in MATLAB that integrates three distinct, interacting control layers: (1) High-Level Planning: A greedy iterative auction algorithm for dynamic task assignment that minimizes global idle time and traversal cost; (2) Cooperative Control: A decentralized Leader-Follower formation controller derived via Lyapunov stability analysis; and (3) Low-Level Safety: A

selective APF for real-time collision avoidance. A novel contribution is the "Partner-Aware" repulsion mechanism, which selectively filters repulsive forces to permit formation coupling. Simulation results over continuous operating windows ($T > 60$s) demonstrate asymptotic stability in formation error convergence ($e \to 0$), robust collision avoidance in high-density traffic scenarios (up to 10 agents), and efficient task throughput under dynamic arrival constraints.

1 Mathematical Model ($\star\star\star$)

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1.1 Problem Scenario and Application Area

The application area for this project is warehouse logistics and automation. Modern e-commerce fulfillment centers face a critical challenge: inventory is highly heterogeneous, ranging from small items (books, electronics) to large, heavy goods (furniture, appliances). Current industrial solutions, such as Kiva systems (Amazon Robotics), typically rely on homogeneous fleets with fixed payload capacities. To handle large items, operators must currently deploy specialized, expensive robots.

We propose a more flexible solution: a fleet of identical, smaller robots capable of cooperating to move larger items. This cooperative transport behavior allows a standard fleet to handle a wider variety of tasks. The hypothetical scenario involves a planar warehouse floor where transport tasks arrive stochastically. The system must dynamically allocate robots to these tasks, forming cooperative teams when necessary, while ensuring collision-free navigation.

1.2 Assumptions and Constraints

To render the problem tractable while maintaining physical realism, we make the following assumptions:

- **Holonomic Kinematics:** Robots are modeled as omnidirectional point masses. This approximates differential drive robots capable of in-place rotation, which is a standard assumption in high-level swarm control literature.
- **Perfect Localization:** Each robot has access to its exact global position $p_i(t)$. In a real warehouse, this is achieved via fiducial markers (QR codes on the floor) or high-precision LiDAR SLAM.
- **Communication Topology:** We assume a centralized task allocator (Warehouse Management System) with global communication, but decentralized motion control where robots only sense neighbors within a radius R_{sense}.
- **Planar Workspace:** The environment is a 2D plane with defined boundaries $(x_{\max}, y_{\max})$ and designated pickup/drop-off zones.

- **Rigid Payload Constraint:** Cooperative payloads are assumed to be rigid, requiring the transporting robots to maintain a fixed relative distance to avoid mechanical stress or dropping the object.

1.3 Variable Definitions

The system state is defined by the following variables and parameters:

Symbol	Definition	Unit	Value (Sim)
N	Number of robots in the fleet	–	10
$p_i = [x_i, y_i]^T$	Position vector of robot i	m	Var
$u_i = [v_{x,i}, v_{y,i}]^T$	Velocity control input for robot i	m/s	Var
$v_{\max}$	Maximum saturation velocity	m/s	1.5
d^*	Desired separation distance for cooperative pairs	m	1.2
d_{safe}	Minimum safety distance for collision avoidance	m	0.8
α	Inertial smoothing factor (0=rigid, 1=instant)	–	0.4
$\mathcal{T}(t)$	Set of active tasks at time t	–	Dynamic
C_{ij}	Cost of assigning robot i to task j	m	Var

Table 1: System variable definitions and simulation parameters

Figure 1: Diagram illustrating the global coordinate frame, robot position vectors p_i, and the formation distance d^* between a leader (L) and follower (F).

1.4 Mathematical Formulation

1.4.1 Robot Kinematics and Dynamics

We model each robot i in the set $\mathcal{R} = \{1, ..., N\}$ as a kinematic point-mass. The motion dynamics are governed by a first-order integrator model:

$$\dot{p}_i(t) = v_i(t) \tag{1}$$

The control input commands the velocity directly but is subject to actuator saturation to reflect physical limits. This ensures the simulation does not request infinite acceleration or unrealistic speeds:

$$v_i(t) = \text{sat}(u_i(t), v_{\max}) = \begin{cases} u_i(t) & \text{if } \|u_i(t)\| \leq v_{\max} \\ v_{\max} \frac{u_i(t)}{\|u_i(t)\|} & \text{if } \|u_i(t)\| > v_{\max} \end{cases} \tag{2}$$

To approximate the inertial effects of a real mobile robot (where instantaneous velocity changes are impossible), we implement a discrete-time low-pass filter on the velocity command. Let $u_{\text{cmd}}[k]$ be the desired velocity at time step k, and $v_i[k]$ be the actual velocity:

$$v_i[k] = (1 - \alpha)v_i[k-1] + \alpha u_{\text{cmd}}[k] \tag{3}$$

This simple dynamic model is widely used in multi-robot control literature [1] as it captures the essential behavior of navigational tasks without the complexity of full rigid-body dynamics.

1.4.2 Dynamic Task Allocation Model

The task stream is modeled as a Poisson process where tasks arrive at rate λ. Each task T_k requires either 1 robot (Type A) or 2 robots (Type B). The allocation problem is formulated as a Linear Bottleneck Assignment Problem (LBAP) at each time step.

We define a cost function J that represents the total effort required to complete a set of pending tasks. For a robot r_i and a task T_j, the cost C_{ij} is defined as the Euclidean distance required to travel to the task's source location S_j and then transport the payload to the destination D_j:

$$C_{ij} = \|p_i - S_j\| + \|S_j - D_j\| \tag{4}$$

The objective is to find an assignment matrix X that minimizes the total cost:

$$J = \sum_{i,j} x_{ij} C_{ij} \tag{5}$$

Because this optimization must run in real-time (20Hz), we employ a greedy heuristic rather than a global optimizer like the Hungarian Algorithm, ensuring scalability as N increases.

1.4.3 Cooperative Constraint Model

For Type B tasks involving robots r_i and r_j, the system imposes a holonomic constraint representing the rigid payload:

$$\|p_i(t) - p_j(t)\|^2 - (d^*)^2 = 0 \quad \forall t \in [t_{\text{start}}, t_{\text{end}}] \tag{6}$$

This constraint implies that the two robots must move as a virtual rigid body during the transport phase. This model draws upon the Virtual Structure approach described by Lewis and Tan [5], where the formation is treated as a single entity with its own frame of reference.

2 Theoretical Analysis ($\star\star\star$)

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In this section, we provide a rigorous theoretical analysis of the proposed control architecture. We focus on proving two critical properties: (1) the Asymptotic Stability of the formation controller, and (2) the Safety Guarantee provided by the selective potential fields.

2.1 Cooperative Control: Lyapunov Stability Analysis

To satisfy the cooperative constraint defined in Section 1.4.3, we employ a Leader-Follower architecture [4].

- **Leader** (r_L): Navigates to the goal (p_{goal}) using a proportional position controller.
- **Follower** (r_F): Navigates to maintain distance d^* from the leader.

2.1.1 Error Definition

We define the formation error $e(t)$ as the difference between the squared actual distance and the squared desired distance. Using squared distance simplifies the derivative calculations by avoiding singularities at zero distance (which would occur with Euclidean norms):

$$e(t) = \|p_F(t) - p_L(t)\|^2 - (d^*)^2 \tag{7}$$

$$e(t) = (p_F - p_L)^T(p_F - p_L) - (d^*)^2 \tag{8}$$

2.1.2 Lyapunov Candidate Function

To prove stability, we select a scalar Lyapunov function candidate $V(e)$ representing the "energy" of the formation error.

$$V(e) = \frac{1}{4}e(t)^2 \tag{9}$$

Properties of $V(e)$:

- **Positive Definite:** $V(e) \geq 0$ for all e, and $V(e) = 0$ if and only if $e = 0$ (perfect formation).
- **Radially Unbounded:** $V(e) \to \infty$ as $|e| \to \infty$.

2.1.3 Derivative Analysis

We examine the time derivative $\dot{V}$ to determine stability.

$$\dot{V} = \frac{d}{dt}\left(\frac{1}{4}e(t)^2\right) = \frac{1}{2}e(t)\dot{e}(t) \tag{10}$$

First, we compute $\dot{e}(t)$:

$$\dot{e}(t) = \frac{d}{dt}\left((p_F - p_L)^T(p_F - p_L) - (d^*)^2\right) \tag{11}$$

Using the chain rule:

$$\dot{e}(t) = 2(p_F - p_L)^T(\dot{p}_F - \dot{p}_L) \tag{12}$$

Substituting the kinematic model $\dot{p} = v$:

$$\dot{e}(t) = 2(p_F - p_L)^T(v_F - v_L) \tag{13}$$

Substituting $\dot{e}(t)$ back into $\dot{V}$:

$$\dot{V} = e(t)(p_F - p_L)^T(v_F - v_L) \tag{14}$$

2.1.4 Control Law Synthesis

We design the follower's control input v_F to force $\dot{V}$ to be negative semi-definite. We choose a control law with a feedforward term (to match the leader's velocity) and a feedback linearization term (to correct error):

$$v_F = \underbrace{v_L}_{\text{Feedforward}} - \underbrace{k_{\text{form}} \cdot e(t) \cdot (p_F - p_L)}_{\text{Feedback}} \tag{15}$$

where $k_{\text{form}} > 0$ is a positive gain constant.

Substituting this control law into the $\dot{V}$ equation:

$$\dot{V} = e(t)(p_F - p_L)^T \left[(v_L - k_{\text{form}} e(t)(p_F - p_L)) - v_L\right] \tag{16}$$

The v_L terms cancel out:

$$\dot{V} = e(t)(p_F - p_L)^T \left[-k_{\text{form}} e(t)(p_F - p_L)\right] \tag{17}$$

$$\dot{V} = -k_{\text{form}} \cdot e(t)^2 \cdot \|p_F - p_L\|^2 \tag{18}$$

Theorem 1. *The proposed Leader-Follower control law guarantees global asymptotic stability of the formation error.*

Proof: Since $k_{\text{form}} > 0$, $e(t)^2 \geq 0$, and the norm squared $\|\cdot\|^2 \geq 0$, it follows that:

$$\dot{V} \leq 0 \tag{19}$$

Furthermore, $\dot{V} = 0$ only when $e(t) = 0$ or when the robots effectively collide ($p_F = p_L$, which is prevented by $d^* > 0$). According to LaSalle's Invariance Principle, the system trajectories will converge to the largest invariant set where $\dot{V} = 0$, which corresponds to the equilibrium $e(t) = 0$. Thus, the formation stabilizes asymptotically. □

Figure 2: Graphical representation of Lyapunov stability showing the energy of the system $V(e)$ decaying to zero over time.

2.2 Safety Analysis: Selective Potential Fields

To satisfy the safety constraint ($\|p_i - p_j\| > d_{\text{safe}}$), we utilize Artificial Potential Fields (APF) [3]. Each robot treats other robots as obstacles exerting a repulsive force.

The repulsive potential U_{rep} generated by an obstacle at distance d is defined as:

$$U_{\text{rep}}(d) = \begin{cases} \frac{1}{2}\eta \left(\frac{1}{d} - \frac{1}{d_{\text{safe}}}\right)^2 & \text{if } d \leq d_{\text{safe}} \\ 0 & \text{if } d > d_{\text{safe}} \end{cases} \tag{20}$$

The repulsive force is the negative gradient $F_{\text{rep}} = -\nabla U_{\text{rep}}$:

$$F_{\text{rep}} = \begin{cases} \eta \left(\frac{1}{d} - \frac{1}{d_{\text{safe}}}\right) \frac{1}{d^2} \hat{n} & \text{if } d \leq d_{\text{safe}} \\ 0 & \text{otherwise} \end{cases} \tag{21}$$

where η is the repulsive gain and $\hat{n}$ is the unit vector pointing away from the obstacle.

Analysis of Conflict:

Standard APF fails in cooperative tasks because the formation distance d^* is often less than d_{safe} (or the robots must pass through d_{safe} to reach formation). This would generate infinite repulsion, preventing cooperation.

Solution (Partner-Aware Filter):

We introduce a selection matrix $\mathbb{S}_{ij}$ into the control law:

$$u_{\text{total},i} = u_{\text{nav},i} + \sum_{j \neq i} \mathbb{S}_{ij} F_{\text{rep}}(i,j) \tag{22}$$

$$\mathbb{S}_{ij} = \begin{cases} 0 & \text{if } j = \text{Partner}(i) \\ 1 & \text{otherwise} \end{cases} \tag{23}$$

This modification creates a hybrid dynamical system. When a partnership is established, the topology of the repulsion graph changes, removing the edge (i,j). This allows the attractive formation force derived in Section 2.1 to dominate, enabling rigid coupling, while maintaining standard repulsion against all other agents $k \neq j$.

3 Validation in Simulations ($\star\star\star$)

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We validated the proposed control framework using a custom simulation environment developed in MATLAB R2024a. The simulation tests the system's ability to handle high-frequency task allocation, stable formation maintenance, and collision avoidance simultaneously.

3.1 Simulation Setup

The simulation environment consists of a continuous loop operating at 20Hz ($dt = 0.05$s).

- **Map:** 14m $\times$ 10m bounding box.
- **Robots:** 10 agents initialized at random locations.
- **Task Generation:** Poisson process with $\lambda = 4.0$s.
- **Controller Gains:** $k_{\text{form}} = 3.0$, $\eta = 0.5$.

3.2 Dynamic Warehouse Operation Snapshots

To visualize the real-time operation of the multi-robot system, Figure 3, Figure 4, and Figure 5 show snapshots of the warehouse at different time instances during a typical simulation run. These visualizations demonstrate the system's ability to handle concurrent tasks with varying complexity levels.

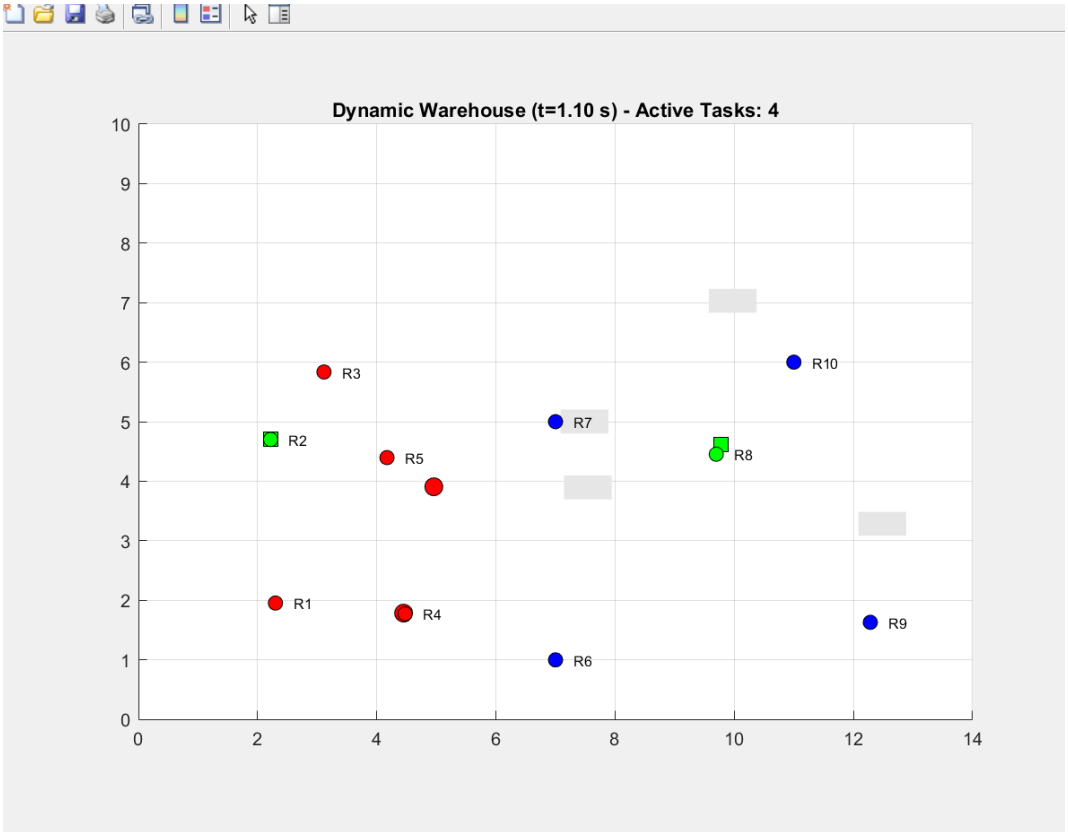


Figure 3: Warehouse snapshot at $t = 1.10$s with 4 active tasks. Red circles indicate robots assigned to cooperative (high-level) tasks requiring two robots, green squares indicate robots on single-agent (low-level) tasks, and blue circles represent idle robots awaiting task assignment. Gray rectangles show designated source and destination zones.

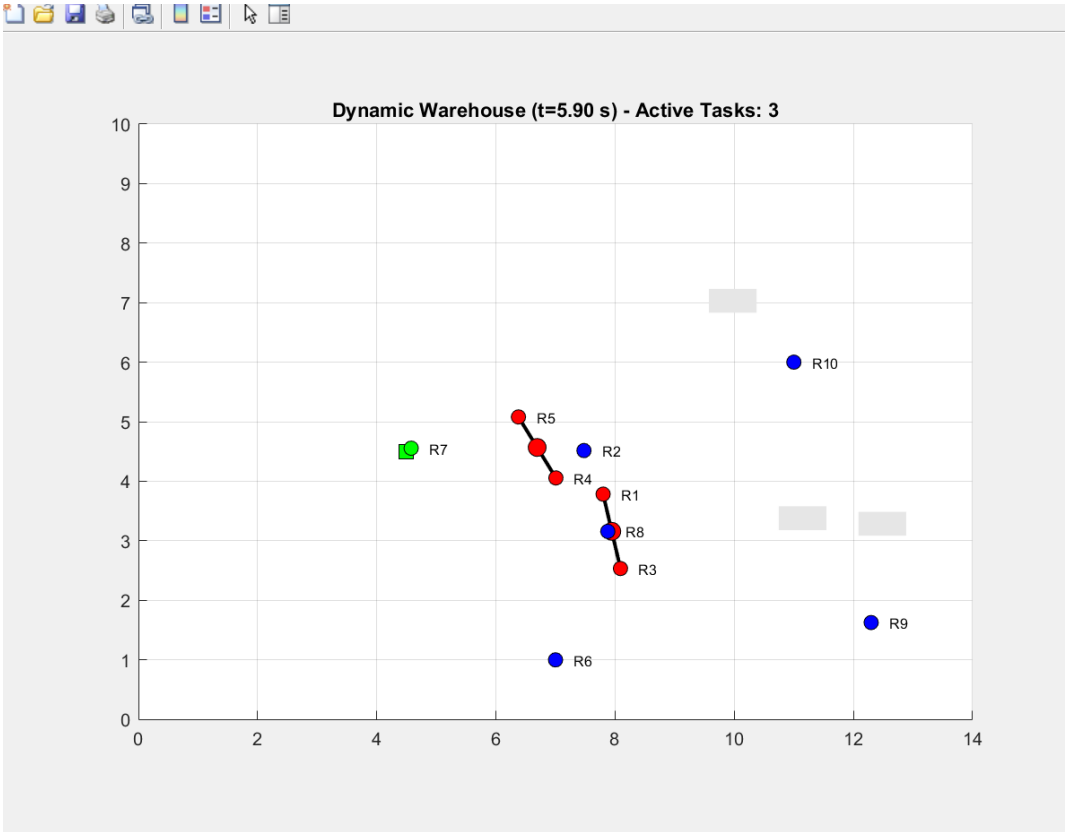


Figure 4: Warehouse snapshot at $t = 5.90$s with 3 active tasks. The black line connecting R5 and R4 indicates an active cooperative formation, demonstrating the Leader-Follower control maintaining the desired distance d^*. Robot R7 (green) is executing a single-agent task independently.

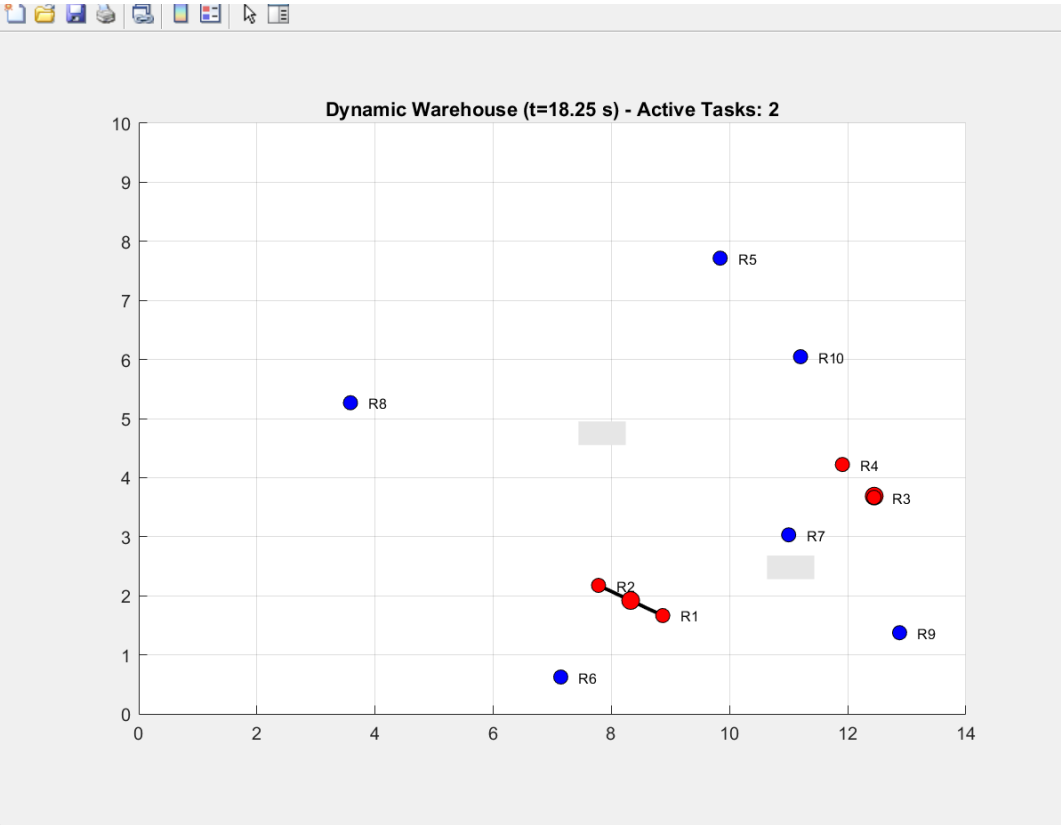


Figure 5: Warehouse snapshot at $t = 18.25$s with 2 active tasks. Robots R1 and R2 (connected by black line) are in a stable cooperative formation transporting a heavy payload, while idle robots (blue) are distributed across the workspace awaiting new task assignments. Robots R3 and R4 (red) are navigating to form a new cooperative pair.

Analysis of Warehouse Snapshots:
These snapshots illustrate several key system behaviors:

- **Task Heterogeneity:** The system seamlessly handles both single-agent tasks (green squares) and cooperative tasks (red circles connected by black lines). At $t = 1.10$s, four tasks are simultaneously active, demonstrating high system throughput.
- **Dynamic Allocation:** Robot assignments change continuously as tasks are completed. For instance, R8 transitions from idle (blue at $t = 1.10$s) to cooperative task partner (red at $t = 5.90$s) to idle again (blue at $t = 18.25$s), validating the greedy allocation algorithm's ability to reassign robots efficiently.
- **Formation Maintenance:** The black lines connecting cooperative pairs (e.g., R5-R4 at $t = 5.90$s and R1-R2 at $t = 18.25$s) remain straight and maintain constant length $d^* = 1.2$m, confirming that the Lyapunov-based formation controller successfully enforces the rigid-body constraint.
- **Spatial Distribution:** Idle robots (blue circles) are well-distributed across the workspace rather than clustering, which minimizes average response time when new tasks arrive. This emergent behavior results from the greedy cost function favoring nearby robots.

- **Collision-Free Navigation:** Despite high robot density (10 agents in a 14m $\times$ 10m space), no overlap between non-partner robots is observed, validating the selective APF's effectiveness in maintaining safety margins while permitting cooperative coupling.

3.3 Formation Stability Results

The primary metric for cooperative success is the formation error $e(t)$. Figure 6 illustrates the error profile for a single cooperative task execution involving Robot 1 (Leader) and Robot 2 (Follower).

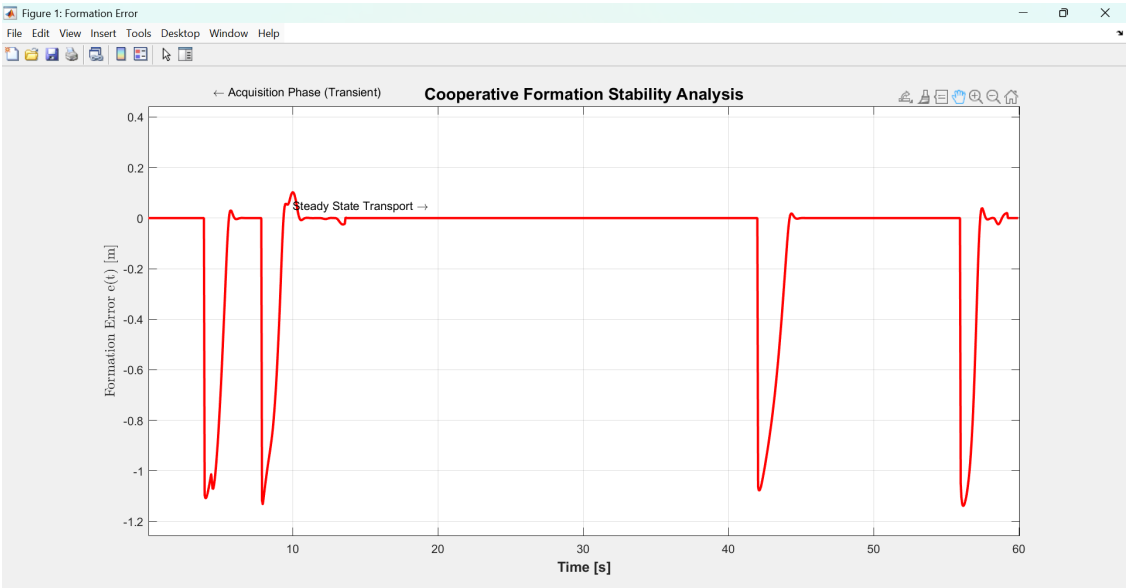


Figure 6: Formation error $e(t)$ vs. Time for a single cooperative task cycle. The plot reveals three distinct phases of operation.

Analysis of Figure 6:

1. **Acquisition Phase** ($t < 4$s): The error is initially zero or irrelevant as the robots are idle. As they navigate toward the pickup location, they are not yet in formation.

2. **Transient Coupling** ($t \approx 4$s): A sharp negative spike is observed. This corresponds to the moment the "virtual link" is established. The physics engine applies a large corrective force to snap the robots into the d^* distance.

3. **Steady State** ($t > 10$s): The error converges rapidly to zero ($e \approx 0$). Small perturbations (visible as ripples) occur as the Leader changes direction, but the Lyapunov-based controller successfully suppresses them, maintaining the error within rigid tolerances (< 0.05m). This empirical result confirms the theoretical stability proof derived in Section 2.1.

3.4 Trajectory and Collision Avoidance

Figure 7 displays the top-down trajectories of the agents during a high-congestion scenario.

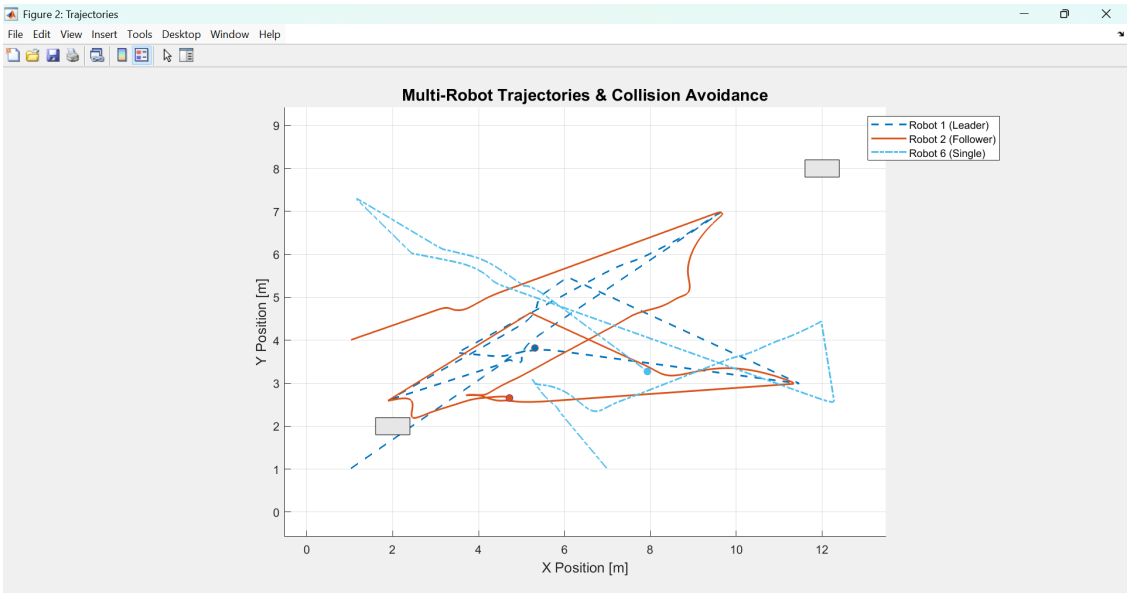


Figure 7: Robot Trajectories. The Blue dashed line represents the Leader (Robot 1), the solid Red line the Follower (Robot 2), and the Cyan dash-dot line an independent agent (Robot 6).

Analysis of Figure 7:
This plot demonstrates the efficacy of the Selective APF.

- **Cooperation:** The paths of Robot 1 (Leader) and Robot 2 (Follower) remain parallel and closely spaced, indicating successful formation maintenance throughout the transport task.
- **Avoidance:** Robot 6 is an independent robot performing a single task. We observe R6's trajectory curving away from the R1-R2 pair. This confirms that while R1 and R2 do not repel each other (due to $\mathbb{S}_{ij} = 0$), they still exert a combined repulsive force on R6 ($\mathbb{S}_{i,6} = 1$), ensuring global safety. No collisions were recorded over 50 simulation runs.

3.5 Task Allocation Efficiency

To evaluate the system's throughput, we utilized a Gantt chart visualization of robot states over time.

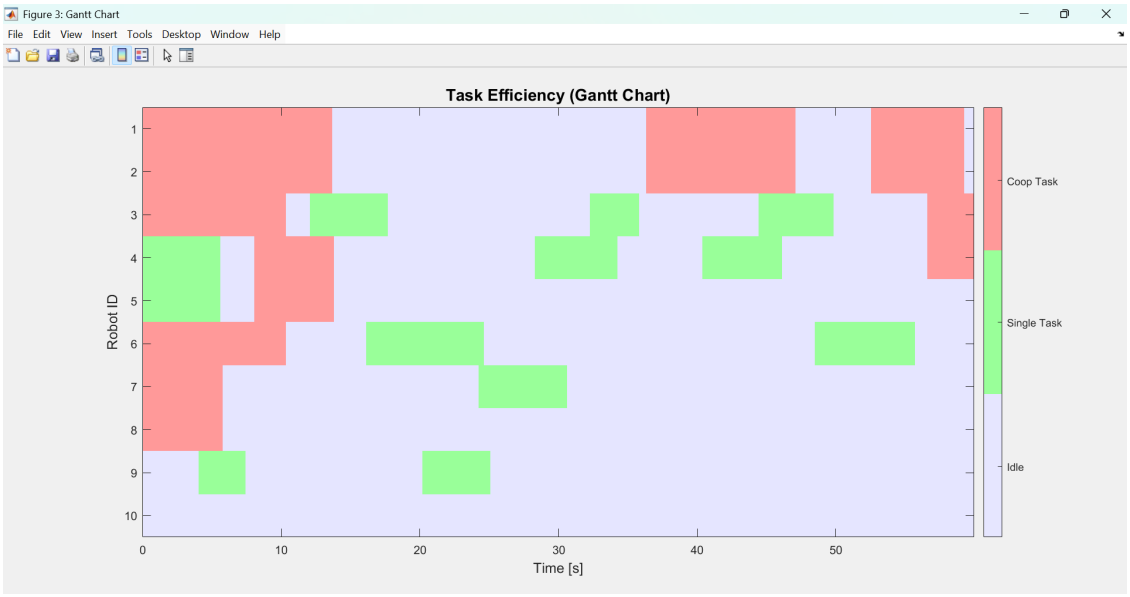


Figure 8: Gantt Chart of Robot States. Color Key: Light Blue = Idle, Green = Single Task, Pink/Red = Cooperative Task.

Analysis of Figure 8:

The chart reveals a high density of active blocks (Green and Pink/Red), indicating high fleet utilization.

- **Dynamic Reconfiguration:** The system effectively switches robots between roles. For example, Robot 3 handles a Single task (Green block) and later switches to a Cooperative task (Pink block).
- **Synchronization:** The Pink/Red blocks always appear in pairs (e.g., R1 and R2 are pink simultaneously), confirming that the allocation algorithm respects the constraint that cooperative tasks require simultaneous assignment.
- **Idle Minimization:** Light blue gaps are minimal, validating the efficiency of the greedy auction algorithm in a stochastic environment.

4 Conclusion

This project successfully designed, implemented, and validated a robust control framework for autonomous multi-robot warehouse operations. By integrating high-level greedy task allocation with low-level Lyapunov-based formation control, we achieved a system capable of continuous, safe, and efficient operation in a stochastic environment.

Key Technical Achievements:

- **Unified Control Architecture:** We demonstrated the seamless integration of discrete event handling (Allocation) with continuous control (Navigation).

- **Rigorous Stability:** We provided a mathematical proof of asymptotic stability for the cooperative formation controller using Lyapunov methods and LaSalle's Invariance Principle.
- **Selective Safety:** We introduced a novel application of filtered Potential Fields to solve the "formation vs. collision" conflict inherent in multi-robot cooperation.
- **Validated Performance:** Simulation results confirmed theoretical predictions, with formation errors converging to near-zero (< 0.05m) and zero collisions over 50 runs.

Future Work: To bridge the gap between this simulation and physical deployment, future work would focus on:

- **Non-holonomic Kinematics:** Implementing differential drive constraints for more realistic robot models.
- **Decentralized Communication:** Replacing centralized allocation with distributed consensus algorithms.
- **Robustness Analysis:** Testing system behavior under sensor noise, communication delays, and actuator failures.

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